3.4.11	**Structure Determination**	
	Data sources	be able to use data from all the analytical techniques listed below to determine the structure of specified compounds
	Mass spectrometry	understand that the fragmentation of a molecular ion $M^{+\bullet} \rightarrow X^{+} + Y^{\bullet}$ gives rise to a characteristic relative abundance spectrum that may give information about the structure of the molecule (rearrangement processes not required)
		know that the more stable X^{+} species give higher peaks, limited to carbocation and acylium (RCO^{+}) ions
	Infrared spectroscopy	be able to use spectra to identify functional groups in this specification
	Nuclear magnetic resonance spectroscopy	understand that nuclear magnetic resonance gives information about the position of ^{13}C or ^{1}H atoms in a molecule
		understand that ^{13}C n.m.r. gives a simpler sprectrum than ^{1}H n.m.r.
		know the use of the δ scale for recording chemical shift
		understand that chemical shift depends on the molecular environment
		understand how integrated spectra indicate the relative numbers of ^{1}H atoms in different environments
		understand that ^{1}H n.m.r. spectra are obtained using samples dissolved in proton-free solvents (e.g. deuterated solvents and CCl_4)
		understand why tetramethylsilane (TMS) is used as a standard
		be able to use the n +1 rule to deduce the spin–spin splitting patterns of adjacent, non-equivalent protons, limited to doublet, triplet and quartet formation in simple aliphatic compounds
	Chromatography	know that gas-liquid chromatography can be used to separate mixtures of volatile liquids
		know that separation by column chromatography depends on the balance between solubility in the moving phase and retention in the stationary phase